

| Question | Answer          | Mark | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                       |
|----------|-----------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14 (a)   | No, with reason | C1   | <p>for No and reason<br/> <b>Acceptable examples</b><br/>           No, when <math>r</math> is divided by 2, <math>m</math> is not multiplied by 2<br/>           No, <math>2 \times 20 = 40</math> but <math>6 \times 16 = 96</math><br/>           No, when <math>m</math> is multiplied by a number <math>r</math> should be divided by the same number<br/>           No, if <math>r</math> is inversely proportional to <math>m</math> then <math>m \times r</math> is constant</p> <p><b>Not acceptable examples</b><br/>           No, <math>m</math> is multiplied by 3<br/>           Yes, he is correct, when <math>m</math> goes up <math>r</math> goes down</p> | <p>Condone the use of '<math>\propto</math>' instead of '<math>=</math>' for the first two M marks</p> <p>Relationship may be implied by substitution</p> |
| (b)      | 200, 8, 10      | M1   | <p>for setting up a proportional relationship between <math>x</math> and <math>y</math>,</p> $y \propto \frac{1}{x^2} \quad \text{or} \quad y = \frac{k}{x^2}$ <p>eg <math>50 \times 2^2 (= 200)</math><br/> <b>OR</b> for <math>50 \times 2^2 (= 200)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                           |
|          |                 | M1   | <p>for process to substitute at least 2 values,</p> $\frac{k}{2^2}$ <p>eg <math>50 = \frac{k}{2^2} \quad (k = 200)</math><br/> <b>OR</b> <math>50 \times 2^2 = y \times 1^2</math> oe <b>or</b> <math>50 \times 2^2 = y \times 5^2</math> oe <b>or</b> <math>50 \times 2^2 = 2 \times x^2</math> oe</p>                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                           |
|          |                 | A2   | for all 3 values correct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                           |
|          |                 | (A1  | for 1 correct value)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 correct value gets M2A1                                                                                                                                 |